

An Internship Report
Of
Java Programming Virtual Internship

CodSoft
Submitted

To

School of Computer Science and Engineering IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410031431

Name of Student: ARYAN SINGH

Batch: 2024-2028

CANDIDATE’S DECLARATION

I, ARYAN SINGH, do hereby declare that the internship report titled “Java Programming Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: ARYAN SINGH
Roll No.: 2410031431

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the CodSoft team for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Java Programming through project-based tasks. I am also thankful to my institute, IILM UNIVERSITY GREATER NOIDA and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: ARYAN SINGH

Roll No. : 2410031431

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



INTERNSHIP OFFER LETTER

Date : 15/07/2026

ID:BY26RY220562

Dear,

Aryan Singh

We would like to congratulate you on being selected for the "Java Programming" virtual internship position with "CodSoft". We at CodSoft are excited that you will join our team.

The duration of the internship will be of **1 Month**, starting from **15 July 2026 to 15 August 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects.

As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Thank You!

Founder



MSME Registered

www.codsoft.in
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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 4-week Virtual Internship undertaken with **CodSoft** in the domain of “Java Programming”. The internship was carried out from 15 July 2026 to 15 August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM UNIVERSITY. The primary aim of this internship was to strengthen core Java programming and object-oriented design skills by building complete, working console and GUI applications that turn theoretical knowledge into practical expertise.

The internship followed a project-based curriculum in which each participant was required to complete a minimum of three tasks out of the five tasks offered for successful completion. The tasks covered a Number Game, a Student Grade Calculator, an ATM Interface, a Quiz Application with Timer, and a Student Course Registration System. All work was maintained in a dedicated GitHub repository, and a short demonstration video of each task was prepared for submission and shared on LinkedIn with the hashtag #codsoft.

4.2 Organization Profile

CodSoft is an innovative tech education and software development platform designed to empower aspiring developers. Its goal is to transform theoretical knowledge into practical expertise. Through virtual internships, AI-driven career tools, and project-based learning curriculums, CodSoft provides a holistic ecosystem for growth, whether a learner is building a first React application or optimizing a complex backend service.

CodSoft believes that the tech industry thrives on continuous learning, and for this reason it created CodSoft AI to act as a round-the-clock mentor, resume reviewer, and technical guide. Participants of its internships also gain access to career tools such as a Resume Builder, ATS Score Checker, AI Interview Coach, Career Roadmap Generator, Portfolio Builder, one-on-one sessions, and interview preparation resources. On successful completion, interns receive a completion certificate, networking opportunities, and placement support. CodSoft is an ISO 9001:2015 certified and MSME-registered organization, working under the tagline “Code.Create.Succeed”. More information is available at www.codsoft.in.

4.3 Problem Statement

Many engineering students learn Java syntax and the principles of object-oriented programming during their coursework, but relatively few get to apply these concepts to complete, end-to-end programs. Topics such as input validation, control flow, class design,

encapsulation, collections, timers, and data management are often understood only in isolation, with little practice in combining them into a usable application.

The problem addressed by this internship was to bridge that gap by designing, coding, testing, and documenting a series of practical Java applications, each with clearly defined requirements, so that theoretical concepts are reinforced through hands-on implementation.

4.4 Project Objectives

- To build a number guessing game in Java using random number generation, loops, conditional logic, attempt limits, multiple rounds, and score tracking.
- To develop a Student Grade Calculator that accepts marks in multiple subjects, computes the total and average percentage, and assigns a grade.
- To design an ATM interface using object-oriented principles, with separate ATM and bank-account classes supporting withdraw, deposit, and balance-check operations with input validation.
- To create a Quiz Application with a per-question timer, multiple-choice questions, score calculation, and a result summary.
- To implement a Student Course Registration System with course and student databases, course listing, registration, and course removal.
- To follow good software practices, including clean code structure, meaningful error messages, version control with Git and GitHub, and demonstration through video.

4.5 Scope of the Project

The scope of this internship covers the design and implementation of Java applications ranging from simple console programs to structured object-oriented systems. It includes taking user input, validating it, performing calculations, managing application state, working with collections of objects, and implementing time-bound behaviour using timers. Each task is self-contained and can be run independently.

The internship was limited to a virtual learning environment. The applications were developed and tested locally as learning projects using in-memory data, and did not involve deployment on any live organizational production system or the use of an external database.

4.6 Technologies and Tools Used

- **Programming Language:** Java (Java Development Kit)
- **Core Libraries:** java.util (Scanner, Random, ArrayList, HashMap), java.util.Timer / TimerTask for the quiz timer

- **Programming Concepts:** Object-oriented programming (classes, objects, encapsulation), control structures, loops, methods, exception handling, input validation
- **Development Environment:** A Java IDE such as Eclipse, IntelliJ IDEA, or VS Code, along with the command-line Java compiler
- **Version Control:** Git and GitHub (repository named CODSOFT_TASKSNO)
- **Presentation:** Screen-recorded demo videos shared on LinkedIn

4.7 System Architecture

Although each task is a separate program, all of them follow a similar layered structure, summarized below.

- **Presentation Layer:** The console (or GUI) menu through which the user enters input and views messages, results, and feedback.
- **Input Validation Layer:** Checks that entered values are within acceptable limits, such as marks between 0 and 100, guesses within the number range, and withdrawals not exceeding the balance.
- **Business Logic Layer:** Core logic of each application, such as comparing guesses, calculating averages and grades, processing ATM transactions, scoring quiz answers, and handling registration rules.
- **Data Model Layer:** Classes such as BankAccount, Question, Course, and Student that store and manage application data using variables and collections.
- **Control Layer:** Classes such as ATM, Quiz, and RegistrationSystem that connect the user interface to the data model and coordinate the overall flow of the program.

4.8 Methodology

The internship followed a task-based methodology in which each task was analysed, designed, implemented, tested, and documented before moving to the next, as summarized in the table below.

Task	Module	Description
1	Number Game	Generates a random number between 1 and 100 and prompts the user to guess it. Gives feedback (correct, too high, or too low), limits the number of attempts, supports multiple rounds, and displays the score.
2	Student Grade Calculator	Takes marks (out of 100) in each subject, calculates total marks and average percentage, assigns a grade based on the average, and displays the results.

Task	Module	Description
3	ATM Interface	Implements an ATM class and a BankAccount class with withdraw(amount), deposit(amount), and checkBalance() methods, validates input, and displays success or failure messages.
4	Quiz Application with Timer	Stores questions, options, and answers; presents one question at a time with a time limit; tracks the score; and displays a result summary of correct and incorrect answers.
5	Student Course Registration System	Maintains course and student data, lists courses with available slots, allows students to register for courses, and allows them to drop registered courses.

Note: There was no stipend associated with this internship.

For every task, the source code was committed to the GitHub repository and a short video demonstrating the working of the application was created. Successful completion required at least three tasks to be submitted through the task submission form shared by CodSoft.

4.9 Expected Outcomes

- Ability to write clear, structured Java programs using loops, conditionals, methods, and user input handling.
- Practical understanding of object-oriented design, including classes, objects, encapsulation, and interaction between multiple classes.
- Skill in validating user input and handling error conditions gracefully.
- Familiarity with Java collections and timer-based logic for building interactive applications.
- Experience in using Git and GitHub to maintain and share project work, and in presenting work through demo videos.
- Successful completion of the required tasks, resulting in the CodSoft Certificate of Completion (C.ID: e31e505) issued on 16 August 2026.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by CodSoft, confirming successful completion of the 4-week Java Programming Virtual Internship (15/07/2026 to 15/08/2026), are attached in Chapter 3 of this report. The offer letter carries ID BY26RY220562 and the certificate carries C.ID e31e505. The GitHub repository link and the LinkedIn post carrying the task demonstration video are attached below as further proof of work.

GitHub Repository: https://github.com/aryansingh-2006/CODSOFT_TASKSNO-.git

5. BIBLIOGRAPHY/REFERENCES

1. CodSoft, “Java Development Internship Task Brief,” www.codsoft.in.
2. CodSoft, “Certificate of Completion – Java Programming,” C.ID: e31e505, dated 16 August 2026.
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